A COHOMOLOGY GROUP OF A PSEUDO-COHERENT COMPLEX CAN BE ISOLATED AS A FREE DIRECT SUMMAND AFTER LOCALIZING

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Allowed standard prerequisites (editorial).

- A pseudo-coherent complex whose cohomology in the relevant degrees is finite can be cut in two at a degree after localizing at a suitable element, producing a direct sum decomposition of the truncations. invoked at proof lines 19852-19856 and again at proof lines 19857-19862. Supplies both splittings: the first separates the truncation above the degree from the rest, the second separates the cohomology module in that degree.

Source reference. Original lemma 0A1U: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19744–19744. Invoked twice, at proof lines 19852-19856 for the upper truncation and at proof lines 19857-19861 for the lower one; each application is made to a different object and the conclusions are combined inside the counted span.

Lemma 0.2. *Let R be a ring. Let $\mathfrak{p} \subset R$ be a prime ideal. Let $K^\bullet$ be a pseudo-coherent complex of R-modules. Assume that for some $i \in \mathbf{Z}$ the maps*

$$H^i(K^\bullet) \otimes_R \kappa(\mathfrak{p}) \longrightarrow H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p})) \quad \text{and} \quad H^{i-1}(K^\bullet) \otimes_R \kappa(\mathfrak{p}) \longrightarrow H^{i-1}(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p}))$$

are surjective. Then there exists an $f \in R$, $f \notin \mathfrak{p}$ such that

(1) *$\tau_{\geq i+1}(K^\bullet \otimes_R R_f)$ is a perfect object of $D(R_f)$ with tor amplitude in $[i+1, \infty]$,*
(2) *$H^i(K^\bullet)_f$ is a finite free R_f-module, and*
(3) *there is a canonical direct sum decomposition*

$$K^\bullet \otimes_R R_f \cong \tau_{\leq i-1}(K^\bullet \otimes_R R_f) \oplus H^i(K^\bullet)_f[-i] \oplus \tau_{\geq i+1}(K^\bullet \otimes_R R_f)$$

in $D(R_f)$.

Proof. We get (1) from Lemma 0A1U as well as a splitting $K^\bullet \otimes_R R_f = \tau_{\leq i} K^\bullet \otimes_R R_f \oplus \tau_{\geq i+1} K^\bullet \otimes_R R_f$ in $D(R_f)$. Applying Lemma 0A1U once more to $\tau_{\leq i} K^\bullet \otimes_R R_f$ we obtain (after suitably choosing f) a splitting $\tau_{\leq i} K^\bullet \otimes_R R_f = \tau_{\leq i-1} K^\bullet \otimes_R R_f \oplus H^i(K^\bullet)_f$ in $D(R_f)$ as well as the conclusion that $H^i(K)_f$ is a flat perfect module, i.e., finite projective. □

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